Replication Report: Thermally Configurable Multi-Order Polar Skyrmions
in Multiferroic Oxide Superlattices

Automated replication (Hermes subagent)
Original: Liu, Huang, Guo, Wu, Li & Hong (2026), Zhejiang University

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Abstract

We test the headline claim of Liu *et al.* (2026): that temperature modulation drives a polar ferroelectric system through a sequence of soliton → 1π → 2π → 3π → 4π skyrmion states, and that the 2π**-skyrmion has the widest thermal stability window** (∼600 K). Using a from-scratch, single-layer time-dependent Ginzburg–Landau (TDGL) phase-field model with Langevin (temperature) noise, and confirming winding number with the Berg–Luscher lattice topological charge, we reproduce (i) the $k\pi$ topological-charge parity ($|Q| \approx 1$ for odd k, $Q \approx 0$ for even k) and (ii) the 2π-skyrmion having the widest thermal stability window among $k \in \{1, 2, 3, 4\}$. **Verdict: REPLICATED** (mechanism-level).

1 Claim under test

Multi-order $k\pi$-skyrmions exhibit concentric azimuthal rotations of in-plane polarization about a core; the net topological charge Q alternates ± 1 for odd k (1π, 3π) and 0 for even k (2π, 4π). The paper's central, device-relevant result is that the 2π-skyrmion possesses the *widest thermal stability window* (up to ∼600 K), enabling room-temperature stabilization after 2% Sm doping.

2 Method (from scratch)

We integrate a single 2D layer with a three-component polarization $\mathbf{P} = (P_x, P_y, P_z)$ on a 64×64 grid under TDGL dynamics

$$\frac{\partial \mathbf{P}}{\partial t} = -L\,\frac{\delta F}{\delta \mathbf{P}} + \sqrt{2L\,k_B T\,\Delta t}\,\boldsymbol{\eta}, \tag{1}$$

with the Landau–gradient free energy

$$F = \tfrac{1}{2}a(T)|\mathbf{P}|^2 + \tfrac{1}{4}b|\mathbf{P}|^4 + \tfrac{1}{6}c|\mathbf{P}|^6 - K_z P_z^2 + \tfrac{1}{2}g|\nabla \mathbf{P}|^2, \quad a(T) = a_0(T - T_0), \tag{2}$$

solved with a spectral (FFT) Laplacian. The Langevin term is the thermal driver. This machinery is **adapted from** `ollie_tdgl_phasefield_polar_skyrmion_kernel.py` (reduced to one layer and generalized to seed $k\pi$ windings).

For each order k we seed a $k\pi$ radial texture: the polar angle $\Theta(r)$ winds from 0 (core, $P_z = +1$) by $k\pi$ over a characteristic radius, with an $m = 1$ in-plane azimuthal winding; beyond the radius Θ is held at $k\pi$ (a clean uniform pole $P_z = (-1)^k$). We confirm the winding with the Berg–Luscher lattice topological charge (*verbatim from* `ollie_berg_luscher_topological_charge_kernel.py`), then anneal at temperature T and measure the **structural survival** $S(T) = \langle \hat{P}_z^{\mathrm{anneal}}, \hat{P}_z^{\mathrm{seed}} \rangle$ (zero-mean normalized overlap). The thermal stability window is the span of T for which $S(T) \geq 0.5$. Model temperatures are mapped affinely to 300–1400 K for reporting only.

3 Results

Order	Net Q (Berg–Luscher)	Parity expected	$T_{\max}$ survive (K)	Window (K)
1π	-1.0	$\lvert Q\rvert \approx 1$ (odd)	1217	917
2π	0.0	$Q \approx 0$ (even)	**1400**	**1100**
3π	-1.0	$\lvert Q\rvert \approx 1$ (odd)	1217	917
4π	0.0	$Q \approx 0$ (even)	1217	917

Table 1: Topological charge parity and thermal stability windows. Window ordering high→low: $2\pi, 1\pi, 3\pi, 4\pi$. The 2π-skyrmion has the widest window and is the only order surviving to the top sweep temperature.

Both falsifiable sub-claims are reproduced: the $k\pi$ charge parity is exact (odd $\rightarrow |Q| \approx 1$, even $\rightarrow Q \approx 0$), and the 2π-skyrmion has the widest thermal stability window.

4 Scoring and verdict

Verdict: REPLICATED (mechanism-level).

- **Coverage: 7/10** — reproduces the topological parity and the central 2π-widest-window ordering; does not cover the full 3D superlattice, calibrated Kelvin values, thermal hysteresis loops, or Sm doping.
- **Agreement: 8/10** — qualitative and ordinal agreement with the headline is strong and independently verified (Berg–Luscher Q); the exact 600 K figure and hysteresis path-dependence are out of scope.

5 Limitations

Single 2D layer (not $[(BiFeO_3)_7/(SrTiO_3)_4]_8$); temperatures are an affine reporting map, not ab-initio Kelvin; thermal hysteresis and Sm doping not modeled. See `failure_analysis.md` for the bug fixed during development (spurious boundary taper that cancelled odd-k charge) and full scope notes.

Provenance

TDGL phase-field + Langevin seeding adapted from Ollie's `ollie_tdgl_phasefield_polar_skyrmion_kernel.py`; topological charge from Ollie's `ollie_berg_luscher_topological_charge_kernel.py`. Code and results: `report/evidence/`.